

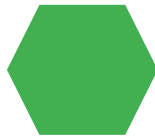


Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

This project analyzes a seasonal agriculture dataset of 4,000 farm records covering 8 crops and 8 states across three cropping seasons - Kharif, Rabi and Zaid. It studies environmental conditions, farming inputs and economic outcomes to understand how agricultural performance changes from season to season.

- Cleaned the data by handling missing values and removing duplicate records
- Compared yield, cost, revenue and profit across the three seasons
- Studied rainfall, temperature, humidity and resource usage season-wise
- Built a correlation heatmap to relate conditions with yield and profit
- Examined irrigation method impact on yield across seasons
- Validated seasonal yield differences with a one-way ANOVA test
- Delivered data-driven insights and recommendations for seasonal planning

WHO ARE THE END USERS?

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- Farmers and agricultural cooperatives - to decide which crops and irrigation methods suit each season
 - Agricultural policymakers and government departments - for evidence-based seasonal planning
 - Agribusiness and agri-tech companies - to guide input supply and pricing decisions
 - Banks and crop insurance providers - to assess seasonal risk and profitability
 - Agricultural researchers and students - to study seasonal patterns and trends
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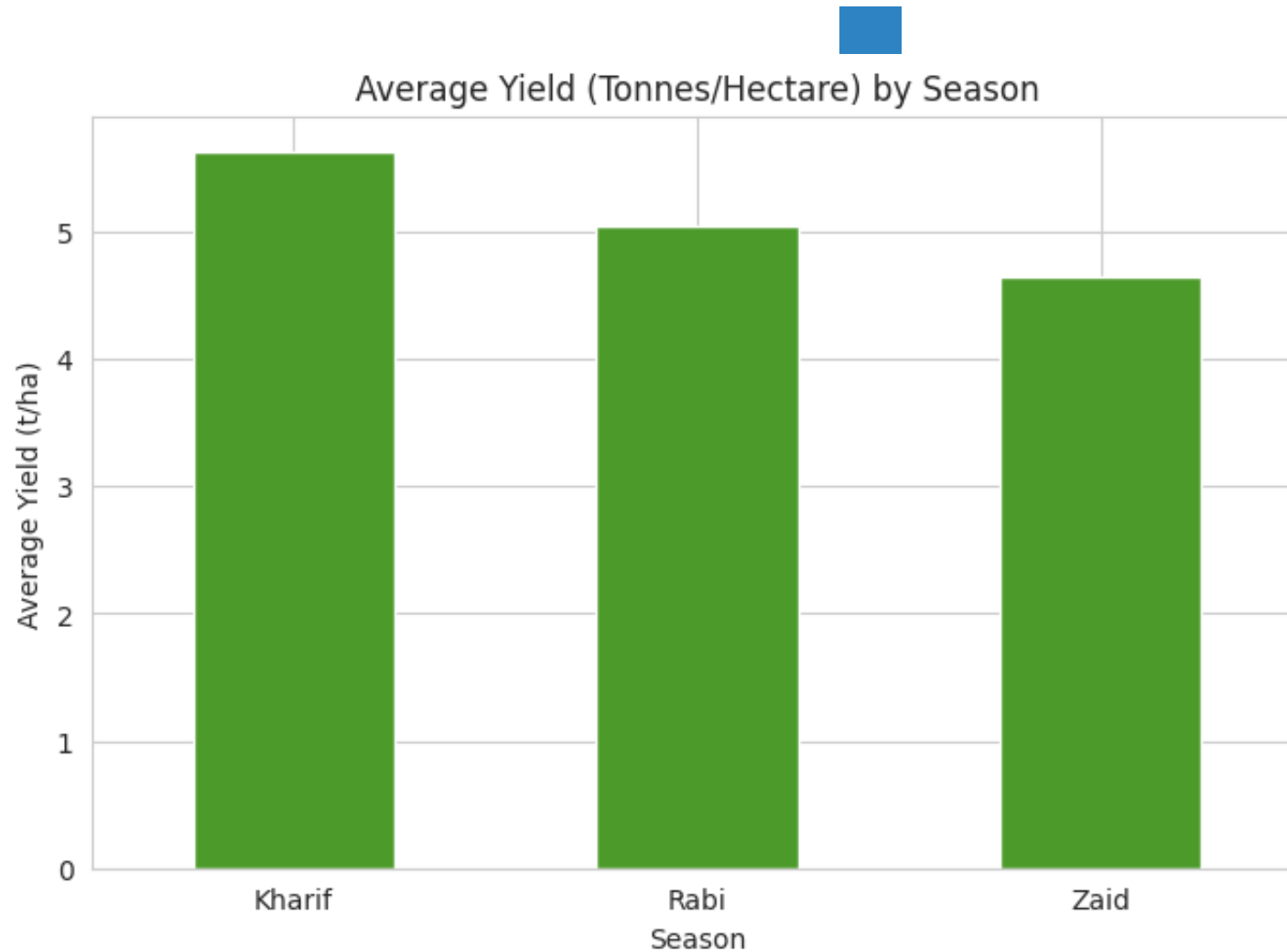
Technology Used

- Python - core programming language for the analysis
- Pandas & NumPy - data cleaning, preparation and analysis
- Matplotlib & Seaborn - charts, boxplots and heatmaps
- SciPy - ANOVA statistical significance testing
- Jupyter Notebook - documenting the complete analysis
- GitHub - version control and project submission



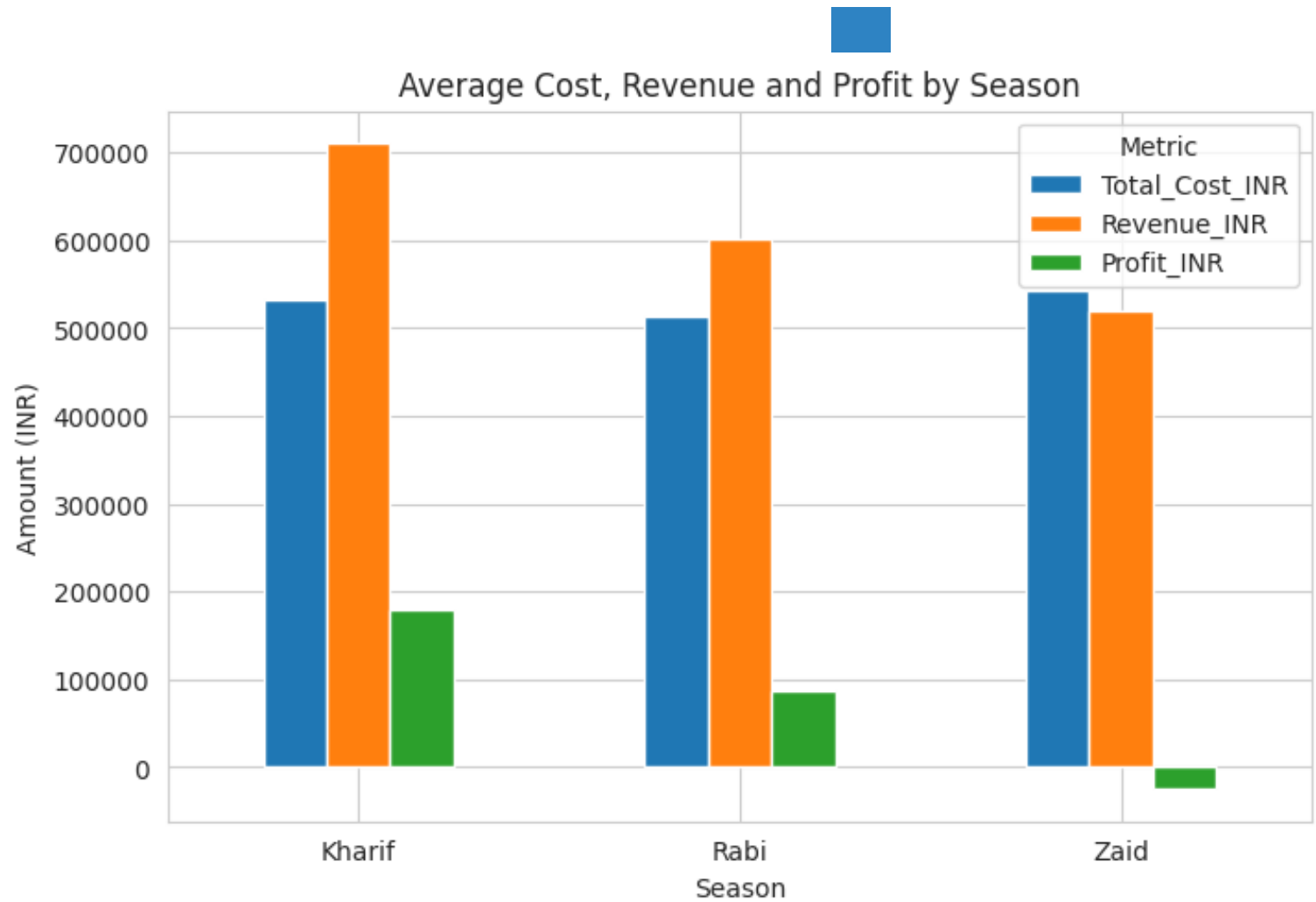
RESULTS

Average Yield by Season



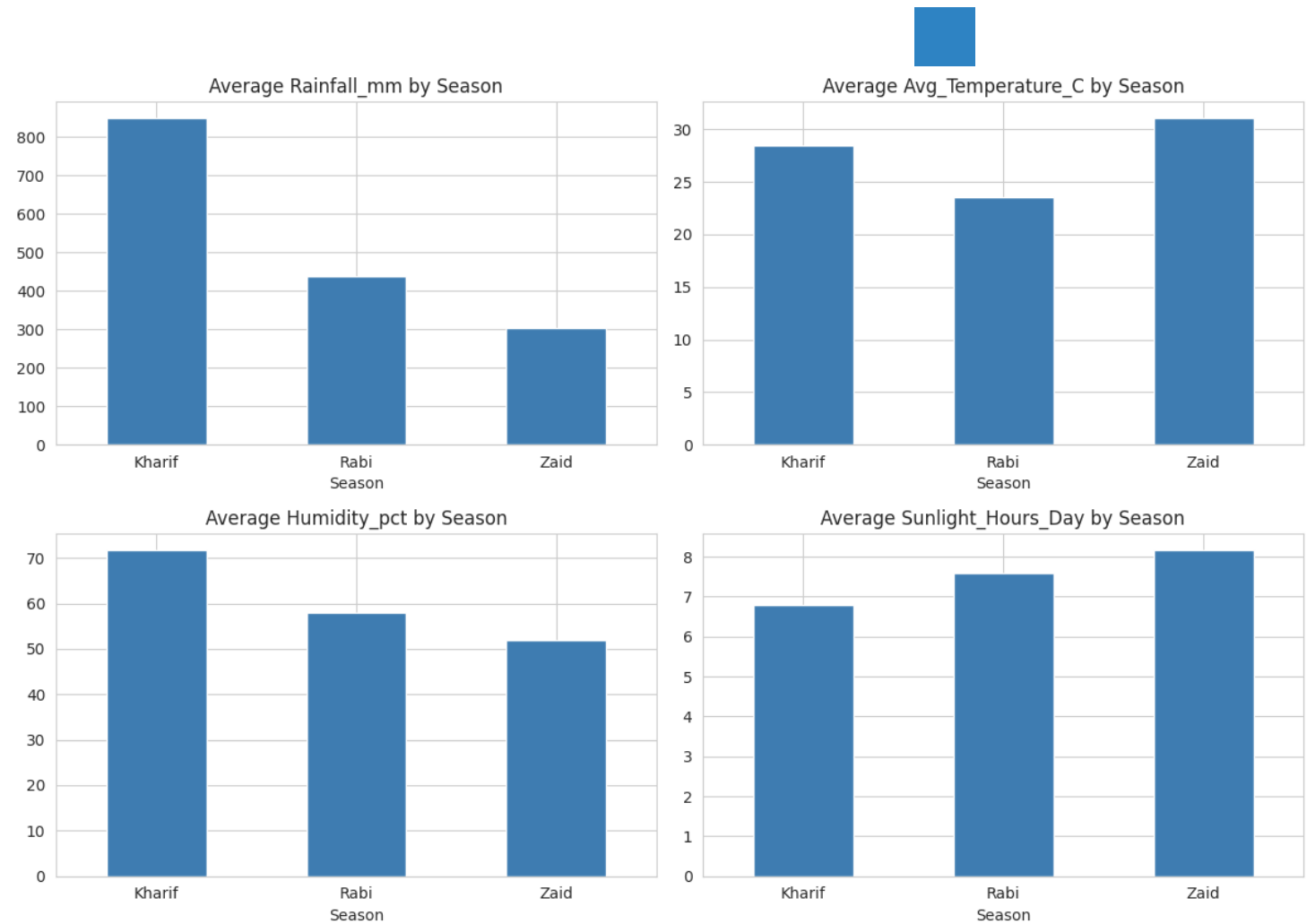
RESULTS

Economic Performance by Season (Cost, Revenue, Profit)



RESULTS

Environmental Conditions by Season



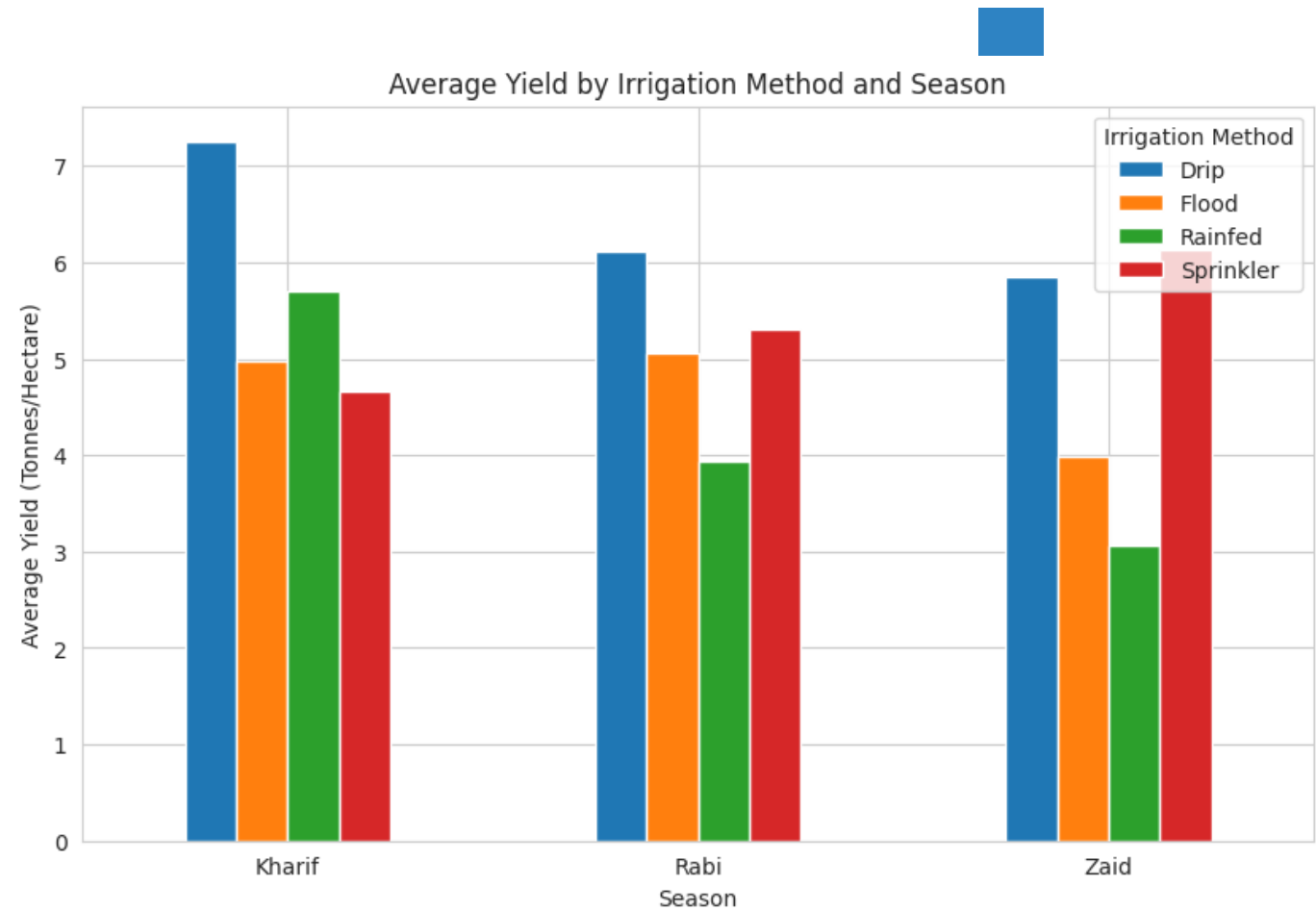
RESULTS

Correlation Between Conditions and Performance



RESULTS

Irrigation Method Impact on Yield by Season



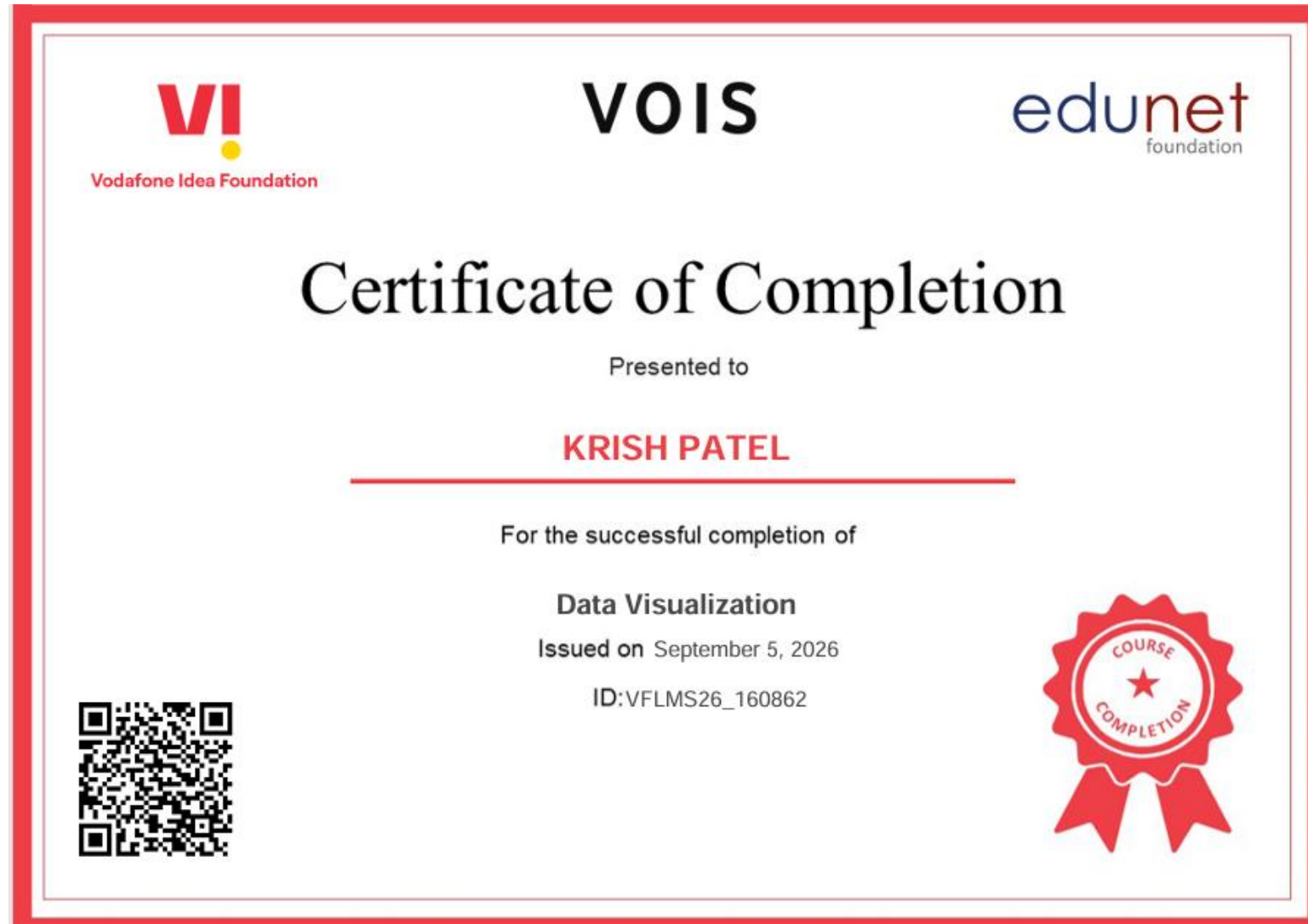
Future scope

- Extend the analysis with multiple years of data to track long-term seasonal trends
- Build a predictive model to forecast yield and profit based on season and weather
- Bring in live weather API data for real-time seasonal monitoring
- Extend the analysis to district or region-level recommendations
- Build an interactive dashboard for farmers and policymakers
- Add market price forecasting to support better economic planning

GitHub Link

[[Krish-p2506/VOIS_Internship](#)]

VOIS Course completion certificate - Course name- Data Visualization



Thank you